Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec:Sr.Super60_NUCLEUS BT | Paper -2(Adv-2022-P2-Model) | Date: 09-11-2025

Time: 02.00Pm to 05.00Pm | GTA-05 | Max. Marks: 180

KEY SHEET

MATHEMATICS

1	**2**	2	**2**	3	**6**	4	**8**	5	**6**	6	**6**
7	**3**	8	**6**	9	**ABC**	10	**AB**	11	**ABD**	12	**ACD**
13	**ACD**	14	**CD**	15	**A**	16	**B**	17	**D**	18	**B**

PHYSICS

19	**5**	20	**2**	21	**0**	22	**4**	23	**5**	24	**0**
25	**3**	26	**4**	27	**ABCD**	28	**ABD**	29	**BCD**	30	**ABC**
31	**ABC**	32	**CD**	33	**C**	34	**D**	35	**A**	36	**A**

CHEMISTRY

37	**7**	38	**7**	39	**9**	40	**6**	41	**1**	42	**7**
43	**7**	44	**6**	45	**BD**	46	**ABCD**	47	**CD**	48	**ABC**
49	**AB**	50	**AB**	51	**D**	52	**B**	53	**A**	54	**D**

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SOLUTIONS

MATHEMATICS

1. n(s) = 5^5 , Favorable cases = $^5C_2$ (5 = 1+1+8 or 5 = 1+2+2) = 1500 P = 12/25

2. $T_n = \cot^{-1}\left[2^{n+1} + \frac{1}{2^n}\right] \Rightarrow T_n = \cot^{-1}\left[\frac{2^{2n+1}+1}{2^n}\right]$

$$T_n = \tan^{-1}\left[\frac{2^n}{1+2^{2n+1}}\right] \Rightarrow T_n = \tan^{-1}\left[\frac{2^{2n+1}-2^n}{1+2^{n+1}.2^n}\right]$$

$$T_n = \tan^{-1}2^{n+1} - \tan^{-1}2^n, S_n = \sum_{n-1}^{n} T_n, S_n = \tan^{-1}\left(2^{n+1}\right) - \tan^{-1}(2)$$

$$S_n = \frac{\pi}{2} - \tan^{-1}2 = \cot^{-1}2$$

3. $\bar{r} = x\bar{a} + y\bar{b} + z\left(\bar{a}\times\bar{b}\right)....(1)$

$$\bar{r}\times\left(\bar{a}\times\bar{b}\right) = \left(\bar{r}.\bar{b}\right)\bar{a} - \left(\bar{r}-\bar{a}\right)\bar{b} \Rightarrow \bar{r}.\bar{b} = 3, \bar{r}.\bar{a} = 2$$

Form equation (1), we get $2 = x\left|a^2\right| + y\bar{a}.\bar{b} \Rightarrow 2x + y = 1$

$$3 = x\bar{a}.\bar{b} + y\left|b\right|^2 \Rightarrow 2x + 4y = 3$$

$$\therefore x = \frac{1}{6}; y = \frac{2}{3}, Also\left|\bar{r}\right| = \frac{4}{\sqrt{3}} \Rightarrow z = \pm\frac{1}{2}, \left|\left[\bar{r}\bar{a}\bar{b}\right]\right| = \left|z\right|\left|a\times b\right|^2$$

4. $a_n a_{n-1} = na_{n-1} + (n-1)a_n, (a_n - n)\left(a_{n-1} - (n-1)\right) = n(n-1)$

$$a_n = \begin{cases} nt + n & if\ n\ is\ odd \\ \frac{n}{t} + n & if\ n\ is\ even \end{cases} where\ t = \frac{20}{23}$$

7. Since $f'(x) = -1 - 3x^2 < 0\ \forall \in R$, Hence f is decreasing

8. $\omega^{23} = 1$

$$N = \sum_{k=1}^{22}\frac{1}{1+\omega^{8k}+\omega^{16k}} = \sum_{k=1}^{22}\left(\frac{1-\omega^{8k}}{1-\omega^{24k}}\right) = \sum_{k=1}^{22}\left(\frac{1-\omega^{8k}}{1-\omega^{k}}\right)$$

$$= \sum_{k=1}^{22}\left[1+\omega^k+\omega^{\omega k}....\omega^{2k}\right]22 + \sum_{k=1}^{22}\left(\omega^k\right) + \sum_{k=1}^{22}\left(\omega^{2k}\right) + ...\sum_{k=1}^{22}\left(\omega^{7k}\right) = 22 - 7 = 15$$

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9. $f(x)=\begin{cases}3-x \text{ of } x<1\\ 2 \text{ of } x=1\\ x+1 \text{ of } x>1\end{cases}$ $g(x) = \min\{f(t) / x \le t \le x^2 + x + 1\}$

$$\begin{cases}2 \text{ of } x \le -1\\ 2-x^2-x \text{ if } -1<x<0\\ 2 \text{ if } 0 \le x \le 1\\ x+1 \text{ if } x>1\end{cases}$$ g(x) is continuous on R

g(x) not differentiable at x = -1, 0, 1

10. Let $P\left(2t_1^2,4t_1\right) and\ Q\left(2t_2^2,4t_2\right)$ ∴ Normals at P and Q intersect on parabola at R(18,−12) which is on the parabola. Which is om the parabola

$\Rightarrow t_1t_2=2t_1+t_2+t_3=0 \quad \therefore t_3\ or\ R\ is-3 \quad \Rightarrow t_1+t_2=3$

11. Clearly, we can see that LHS = $\sqrt{f\left(x^3\right)}$ and RHS = $2+\sqrt[3]{g\left(x^2\right)}$

Thus, we can try to convert the equation into the form of $f\left(x\right)=f^1\left(x\right)$

Adding a constant t to both sides,

$$\sqrt{x^3+3x^2+3x+9}+\left(t\right)=\left(2+t\right)+\sqrt[3]{x^2+6x+1}$$

Taking RHS as f(x) $f^{-1}\left(x\right)=\sqrt{\left(x-2-t\right)^3+8}-3$

Clearly, we can see that putting t = -3 gets us $f^{-1}\left(x\right)=LHS$

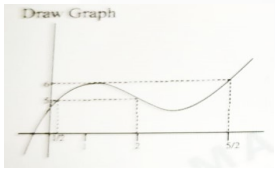
Hence, we can see that putting t = -3 gets us $f^{-1}\left(x\right)=LHS$

Hence, we need to solve: $\sqrt{x^3+3x^2+3x+9}-3=x$

$x^3+3x^2+3x+9=x^2+6x+9, x^3+2x^2-3x=0$

Hence, x = -3, 0, 1 are the solutions.

12.



13. $f\left(x\right)=e^x\sin x\Rightarrow f'\left(x\right)\ \&\ f''\left(x\right)>0\forall x\in\left(0,\frac{\pi}{2}\right)$

Area A_1 is lesser than half of the formed by coordinate axes $x=\frac{\pi}{2}\ \&\ y=f\left(\frac{\pi}{2}\right)$

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15. $\sum_{k=1}^{n}\frac{k^2-\frac{1}{2}}{\left(k^2+\frac{1}{2}\right)^2-k^2}$, $\sum_{k=1}^{n}\frac{k^2-\frac{1}{2}}{\left(k^2+k+\frac{1}{2}\right)\left(k^2-k+\frac{1}{2}\right)}$ $\sum_{k=1}^{n}\left(\frac{k-\frac{1}{2}}{k^2-k+\frac{1}{2}}-\frac{k+\frac{1}{2}}{k^2+k+\frac{1}{2}}\right)$

16. x can be 2, 3, 4, 5, 6
The number of ways in which sum of 2, 3, 4, 5, 6 can occur is given by the coefficients of x^2,x^3,x^4,x^5,x^6 in $(3x + 2x^2 + x^3)(x + 2x^2 + 3x^3) = 3x^2 + 8x^3 + 14x^4 + 8x^5 + 3x^6$
From above it is clear that sum that occurs most often is 4

17. $\overline{a_1}=(a,b),\overline{a_2}=\left(a-2,b-\sqrt{5}\right),\overline{a_3}=\left(a+2,b-\sqrt{5}\right)$

Area of $\triangle$, Whose vertices are $\overline{a_1},\overline{a_2}\ and\ \overline{a_3}$ is $\Delta=\frac{1}{2}\begin{vmatrix}1 & a & b\\ 1 & a-2 & b-\sqrt{5}\\ 1 & a+2 & b-\sqrt{5}\end{vmatrix}$

$$\Delta^2=\frac{1}{4}\begin{vmatrix}1 & a & b\\ 1 & a-2 & b-\sqrt{5}\\ 1 & a+2 & b-\sqrt{5}\end{vmatrix}\begin{vmatrix}1 & a & b\\ 1 & a-2 & b-\sqrt{5}\\ 1 & a+2 & b-\sqrt{5}\end{vmatrix}$$

And $\left|\overline{a_1}-\overline{a_2}\right|=3,\left|\overline{a_1}-\overline{a_3}\right|=3,\left|\overline{a_2}-\overline{a_2}\right|=4$ $\Delta=\sqrt{5\times2\times2\times1}=2\sqrt{5}$

$$\left(2\sqrt{5}\right)^2=\frac{1}{4}\begin{vmatrix}1+\overline{a_1}.\overline{a_1} & 1+\overline{a_1}.\overline{a_2} & 1+\overline{a_1}.\overline{a_3}\\ 1+\overline{a_2}.\overline{a_1} & 1+\overline{a_2}.\overline{a_2} & 1+\overline{a_2}.\overline{a_3}\\ 1+\overline{a_3}.\overline{a_1} & 1+\overline{a_3}.\overline{a_2} & 1+\overline{a_3}.\overline{a_3}\end{vmatrix}$$

Required answer = $4\times5\times4=80$

18. $\sin nx-\sin(n-2)x=2\cos(n-1)x\sin x$

$$\int\frac{\sin nx}{\sin x}dx=\int2\cos(n-1)dx+\int\frac{\sin(n-2)x}{\sin x}dx$$

$$\int_0^{\frac{\pi}{2}}\frac{\sin5x}{\sin x}dx=\int_0^{\frac{\pi}{2}}2\cos4xdx\int_0^{\frac{\pi}{2}}\frac{\sin3x}{\sin x}dx\ =0+\int_0^{\frac{\pi}{2}}\frac{\sin3x}{\sin x}dx\ =\int_0^{\frac{\pi}{2}}dx=\frac{\pi}{2}$$

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PHYSICS

19. KE of car converts into elastic potential energy of springs

$\frac{1}{2}mv^2 = \frac{1}{2}k_1\left(\frac{50}{100}\right)^2 + \frac{1}{2}k_2\left(\frac{20}{100}\right)^2$, Solve to give v=0.5 ms^{-1}

20. $R_1 = \lambda N_1, R_2 = \lambda N_2$, No of atom decayed in $(T_2 - T_1)$

$= N_1 - N_2 = \frac{R_1 - R_2}{\lambda} = \frac{(R_1 - R_2)T}{\ln 2} = \frac{2(R_1 - R_2)T}{\ln 4}$. Hence n = 2

21. $P_0V_0 = nRT_a$

$(P_0 + \Delta P)(V_0 + \Delta V) = nRT_b, (P_0 + \Delta P)(V_0 - \Delta V) = nRT_b, \Delta V = 0$

Pressure depends on number of moles and is independent of nature of gas

22. $E(t) = A^2e^{-\alpha t}$

In E=2ln A- αt $\quad \therefore \frac{\Delta E}{E} \times 100 = 2\frac{\Delta A}{A} \times 100 - \alpha\Delta t \times 100$

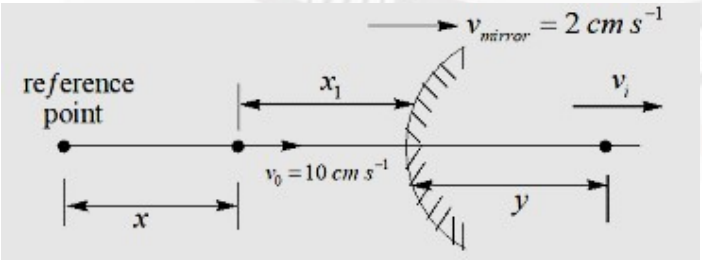
For time $\frac{\Delta t}{t} \times 100 = 1.5$, At $t = 5s. \Delta t = \frac{1.5 \times 5}{100} = \frac{7.5}{100}$

Using eq(i) % error in $E = 2 \times (1.25) + \alpha \times \frac{7.5}{100} \times 100 \quad = 2.5 + 1.5 = 4$

23. Net current $I = 3 + 4\sqrt{2}\sin\omega t$

Now find, $I_v = \sqrt{\frac{\int_0^t \left(3 + 4\sqrt{2}\sin\omega t\right)^2 dt}{T}} = 5A$

24. $\frac{1}{y} + \frac{1}{-x_1} = \frac{1}{10} \quad \frac{1}{y} - \frac{1}{10} = \frac{1}{10} \Rightarrow y = 5cm$



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$$\frac{dx}{dt}=10cm\,s^{-1},\frac{d\left(x+x_1\right)}{dt}=2\,cm\,s^{-1}\,\frac{dx_1}{dt}=-8\,cms^{-1}$$

$$\frac{dy}{dt}=v_i-2\quad form(i)\frac{dy}{dt}=\left(\frac{y}{x_1}\right)^2\frac{dx_i}{dt}$$

$$v_i-2=\left(\frac{5}{10}\right)^2(-8)\Rightarrow v_i=0$$

25. Solid angle subtended by plane surfaces

$$\Omega=2\times2\pi\left(1-\cos\theta\right)=4\pi\left(1-\cos\theta\right)$$

Flux through two plane surface $\phi_0=\frac{Q\Omega}{4\pi\varepsilon_0}=\frac{Q}{\varepsilon_0}\left(1-\cos\theta\right)$

Total flux = $\frac{Q}{\varepsilon_0}$ And flux though curved surface

$$\frac{Q}{\varepsilon_0}-\frac{Q}{\varepsilon_0}\left(1-\cos\theta\right)=\frac{Q}{\varepsilon_0}\cos\theta\quad\therefore\phi=\frac{Q}{\varepsilon_0}\cos\theta$$

For $\theta=30^0,\phi_{30^0}=\frac{\sqrt{3}Q}{2\varepsilon_0}$ ..... (i)

For $\theta=60^0,\phi_{60^0}=\frac{Q}{2\varepsilon_0}$ ..... (ii)

Using eq (i) and (ii)]

$$\frac{\phi_{30^o}}{\phi_{60^o}}=\frac{\sqrt{3}Q}{2\varepsilon_0}\times\frac{2\varepsilon_0}{Q}=\sqrt{3}.So,\frac{\phi}{\phi/\sqrt{n}}=\sqrt{3},n=3$$

26. $\eta=\frac{F}{A\theta}$

Here $A=L^2\,and\,\theta=\frac{x}{L}$

Restoring force is $F=-\eta A\theta=-\eta Lx$

Acceleration is $a=\frac{-\eta l}{m}x$

$a\alpha-x$

$\therefore$ Oscillation are simple harmonic with

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$$\omega = \sqrt{\frac{\eta L}{m}}$$

$$so,\ T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{\eta L}} = \pi\sqrt{\frac{4m}{\eta L}}$$

$$\alpha = 4$$

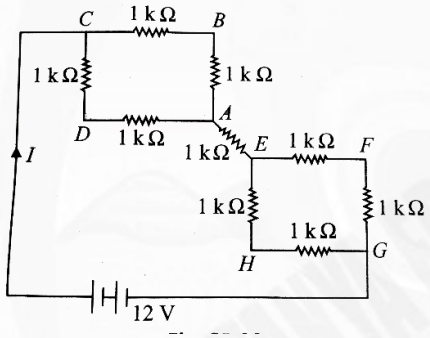
27. A) Only equilibrium position changes Time period remains same $T = 2\pi\sqrt{m/k}$

B) $\frac{1}{2}kx^2 = \frac{1}{2}mv^2$

So v at mean position is same amplitude will be same

C) In case 4 equilibrium position is $x_0 = 3mg/k.$

28. The circuit can be redrawn as shown in the figure



$$R_{eq} = 3k\Omega,\ I = \frac{12}{3\times10^3} = 4mA$$

$$V_E - V_G = I(1k\Omega) = 4\times1\times10^{-3}1\times10^3 = 4V$$

29. Charge on inner sphere can be supposed to be concentrated as a point charge at the centre, hence electric field at a point in the region between the spheres at a distance r from the centre is ($q/4\pi\varepsilon_0 r^2$). Due to induction, equal and opposite charges will appear on the inner and outer surfaces of the outer sphere. Hence, net charge which can be supposed to be placed at the centre is q and electric field due to it at a point outside the hollow sphere at a distance r from centre is ($q/4\pi\varepsilon_0 r^2$).

Potential of inner sphere is $\frac{q}{4\pi\varepsilon_0}\left(\frac{1}{a}-\frac{1}{b}+\frac{1}{c}\right)$

And potential of outer sphere is $\frac{q}{4\pi\varepsilon_0 c}$

Potential of inner sphere with respect to the outer sphere $\frac{q}{4\pi\varepsilon_0}\left(\frac{1}{a}-\frac{1}{b}\right)$

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30. Path difference $\delta = BP - AP$

$$\sqrt{x^2 + 9^2} - x = n\lambda$$

$$\Rightarrow x^2 + 9^2 = n^2\lambda^2 + x^2 + 2n\lambda x$$

$$\Rightarrow x = \frac{9^2 - n^2\lambda^2}{2n\lambda}$$

$$\lambda = \frac{c}{v} = \frac{3\times10^8}{120\times10^6} = 2.5m$$

$$n = 1, x = 14.95m; n = 3, x = 1.65m;$$

$$n = 2, x = 5.6m; n = 4 \text{ not possible}$$

31. Rise water in capillary tube $h = \frac{2\sigma\cos\theta}{\rho rg}$

For a given material of the capillary tube $h\alpha\frac{1}{r}, h\alpha\sigma$

When lift is going up with a constant acceleration a then effective acceleration in the experiment will be a + g

$$h' = \frac{2\sigma\cos\theta}{\rho r(a+g)} \text{ and } h' < h$$

32. Surface tension $= S, \gamma = 5/3$

$$r_1 \to T_1 \to P_{a1}$$

$$r_2 \to T_2 \to P_{a2}$$

For bubble the pressure is $P_{gas} = p_a + \frac{4s}{r}$

As for isothermal condition
PV = constant

So, $\left(p_{a1} + \frac{4s}{r_1}\right)\left(\frac{4}{3}\pi r_1^3\right) = \left(p_{a_2} + \frac{4s}{r_2}\right)\left(\frac{4}{3}\pi r_2^3\right)$

$$\frac{r_1^3}{r_2^3} = \left(\frac{p_{a2} + \frac{4s}{r_2}}{p_{a1} + \frac{4s}{r_1}}\right)$$

Now $P^{1-\gamma}T^{\gamma}$ =Constant

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$$\left(p_{a_2}+\frac{4s}{r_2}\right)^{1-\frac{5}{3}}T_2^{\frac{5}{3}}=\left(p_{a1}+\frac{4s}{r_1}\right)^{1-\frac{5}{3}}T_1^{\frac{5}{3}}$$

$$\left(\frac{T_2}{T_1}\right)^{\frac{5}{3}}=\left(\frac{p_{a_1}+\frac{4s}{r_1}}{p_{a_2}+\frac{4s}{r_2}}\right)^{\frac{-2}{3}}$$

$$\left(\frac{T_2}{T_1}\right)^{\frac{5}{2}}=\left(\frac{p_{a_2}+\frac{4s}{r_2}}{p_{a_1}+\frac{4s}{r_1}}\right)$$

33. Let v be the liner velocity of centre of mass of the spherical body and ω its angular velocity about centre of mass

Then

$$\omega=\frac{v}{2R}$$

K.E of spherical body

$$K_1=\frac{1}{2}mv^2+\frac{1}{2}\omega^2 I$$

$$=\frac{1}{2}mv^2+\frac{1}{2}(2mR)^2\left(\frac{v^2}{4R^2}\right)$$

$$=\frac{3}{4}mv^2$$

Speed of the block will be

$$v'=(\omega)(3R)=3R\omega=(3R)\left(\frac{v}{2R}\right)=\frac{3}{2}v$$

K.E of block $K_2=\frac{1}{2}mv^2=\frac{1}{2}m\left(\frac{3}{2}v\right)^2=\frac{9}{8}mv^2...(ii)$

Form equation (i) and (ii), $\frac{K_1}{K_2}=\frac{2}{3}$

34. Momentum corresponding to incident photons normal to the surface

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$$\left(\frac{dp}{dt}\right)_{\text{incident}} = \frac{I}{c}dA\cos^2\theta$$

Since reflection coefficient is ρ, so the momentum of the reflected photons per second normal to surface

$$\left(\frac{dp}{dt}\right)_{\text{reflected}} = -\frac{I}{c}dA\rho\cos^2\theta$$

Hence, rate of change of momentum of the photons,

$$\left(\frac{dp}{dt}\right)_{\text{photons}} = -\frac{I}{c}dA(\rho+1)\cos^2\theta$$

From Newton's third law,

$$\left(\frac{dp}{dt}\right)_{\text{surface}} = -\frac{I}{c}dA(\rho+1)\cos^2\theta$$

Hence, pressure exerted on surface

$$p = \frac{dF}{dA} = \frac{I}{C}(\rho+1)\cos^2\theta$$

On substituting values, we get P= 0.6nN cm^{-2}

35. Here $v = e^a h^h \mu^c G^d$ taking the dimensions, $M^0LT^{-1}A^0$

$$= \left[AT^1\right]^a \left[ML^2T^{-1}\right]^b \left[MLT^{-2}A^{-2}\right]^c \left[M^{-1}L^3T^{-2}\right]^d$$

There will be four simultaneous equations by equating the dimensions of M, L, T and A. These are

$a-2c=0, a-b-2c-2d=-1, b+c-d=0$ and $2b+c+3d=1$

Solving for a, b, c and d we get

$a=-2, b=1, c=-1, d=0$

Thus $v = e^{-2}h\mu^{-1}G^0$

36. Electric field in between the capacitor plates is given by

$$E = \frac{Q}{\varepsilon_0 A}$$

Where Q is the charge on capacitor

$$Q = \varepsilon_0 A \times E = 8.85\times10^{-12}\times25\times10^{-7}\times400 = 8.85\times10^{-15}C$$

Magnetic force experienced by +ve plate is

$$F_m = QvB = 8.85\times10^{-15}\times25\times4$$

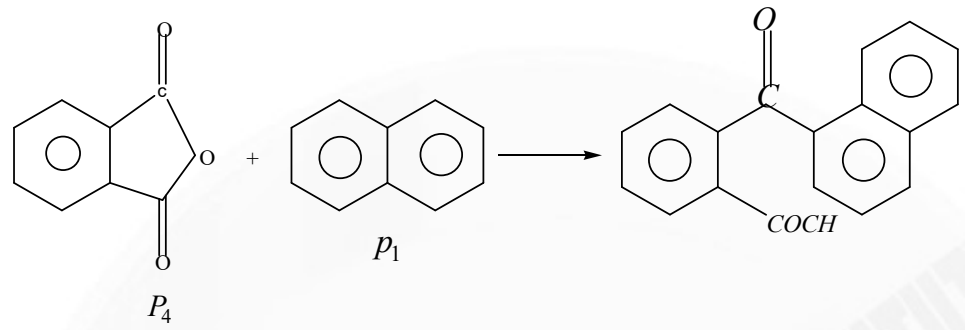
$= 8.85\times10^{-13}N$ In direction out of the plane of paper

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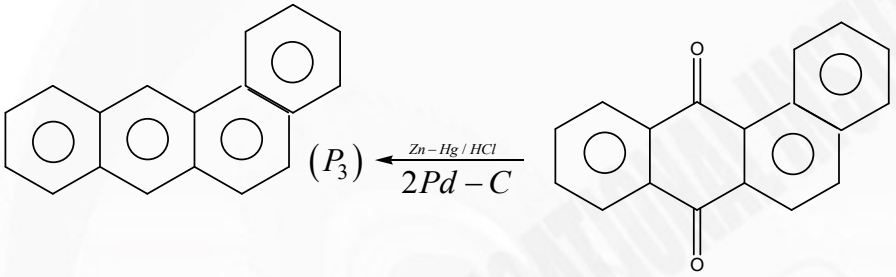
CHEMISTRY

37. Bond order in BF_3 graphite, NO_3^-, CO_3^{2-}, SO_3^{2-} is 1.33. ClO_3^- bond order 1.66, In O_3 bound order is 1. In diamond and carborrudum bonder is 1.

38.



P_4 p_1



X = 13, Y = 6, Z = 12

39. Gas A is O_2 B is O_3 X is I_2 and product is I_4O_9

40. The formula for conductance is $G = K \times \frac{a}{\ell}$

$$5\times10^{-7} = k \times \frac{1}{120}$$

$$k = 6\times10^{-5}\, s\ cm^{-1}$$

$$\Lambda_m^c = \frac{k\times100}{M} = \frac{6\times10^{-5}\times1000}{0.0015} = 40$$

$\because pH = 4$

$$\therefore \left[H^+\right] = 10^{-4} = c\alpha = 0.0015\alpha$$

$$\alpha = \frac{10^{-4}}{0.0015}$$

$$\alpha = \frac{\Lambda_m^c}{\Lambda_m^o} \Rightarrow \frac{10^{-4}}{0.0015} = 40$$

$$\Lambda_m^o = 6\times10^2\, s\, cm^2 mole^{-1}$$

$i.e\ z = 6$

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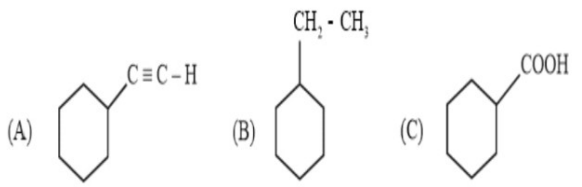
41. Rate law $= k_2\left[N_2O_2\right]\left[H_2\right]$ [$\because$ slowest step reaction is RDS]

$$\because \frac{k_1}{k-1}=\frac{\left[N_2O_2\right]}{\left[NO\right]^2} \because \left[N_2O_2\right]=\frac{k_1}{k-1}\left[NO\right]^2$$

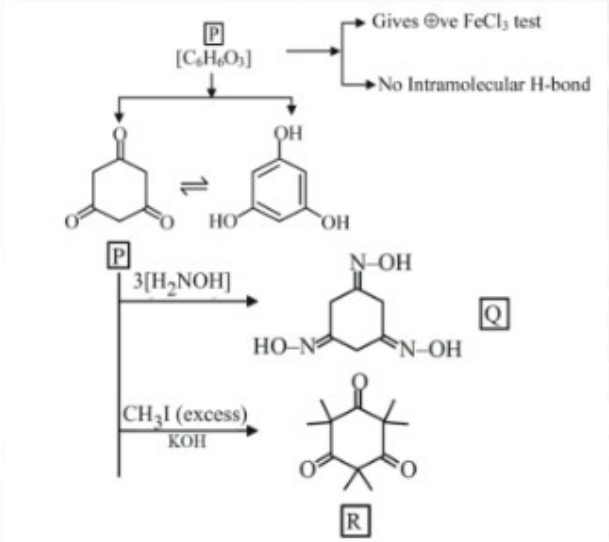
$$\because \text{ Rate} = k_2 \times \frac{k_1}{k-1}\left[NO\right]^2\left[H_2\right]$$

42. (p) is $CrO_2C\ell$; (Q) is Toluene ; (R) is Benzaldehyde

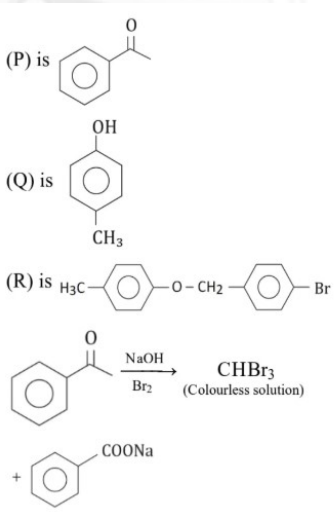
43.



44.

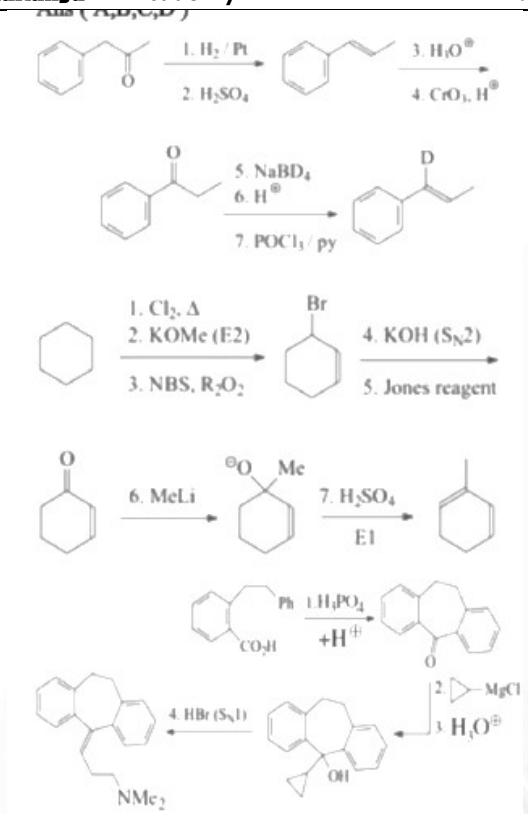


45.



46.

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47. Conceptual

48. $X \rightarrow X^{+2} + 2e^-$

$y^{+2} + 2e^- \rightarrow y_{(s)}$

$$E_{Cell} = E^0_{cell} = \frac{0.06}{2}\log\frac{x^{+2}}{y^{+2}}$$

49. (A) As adsorption is spontaneous, ΔG for the process is – ve. Adsorption is accompanied by decrease in randomness. Therefore ΔS and TΔS for the process is also negative. As ΔS for the process is –ve and the process is spontaneous, ΔH for the process has to be –ve i.e, enthalpy of the system decreases.
(B) Under a given set of conditions of temperature and pressure the easily liquefiable gases e.g. C_2H_6, NH_3 and HCl are adsorbed more than the gases like N_2, H_2 *and* CO. The ease with which a gas can be liquefied is determined by its critical temperature.
Critical temperature is the minimum temperature above which a gas can be liquified. This implies that gases with high critical temperature values can be easily liquified as compared to gases with low critical temperature value.

50. Conceptual
51. Conceptual
52. Conceptual
53. Conceptual
54. Conceptual

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